

Chua Yuan Shan

chuanyuanshan@gmail.com ❖ +65 8787 1368 ❖ GitHub ❖ LinkedIn ❖ Singapore

EDUCATION

Nanyang Technological University

August 2025 – August 2029

Bachelor of Computing (Hons) in Data Science and Artificial Intelligence

GPA: 4.8/5.0, Scholarship: Turing AI Scholars Program (TAISP)

WORK EXPERIENCE

iQiYi International

August 2026– Jan 2027

Full Stack Developer Intern

- Developed an end-to-end automated dubbing pipeline, **streamlining audio preprocessing, model inference, and segment synthesis** to generate natural, coherent speech. Results showed a **67% efficiency gain** over manual work and **reduced 7-10 man-hours** per episode.
- Evaluating use of **Text-to-Speech (TTS)** and **Voice Conversion (VC)** models, fine tuning models (GPT-SoVITS, Qwen-3.0-TTS, Seed-VC, Chatterbox) to tackle the issue of cross-lingual inference.
- Evaluated and fine-tuned **Speech Emotion Recognition (SER)** models, including **emotion2vec** and fine-tuned **wav2vec2**, to extract emotion and timbre-matched reference audio for background character voice synthesis.
- Adding features and debugging for the frontend and backend of the automated dubbing website, working with **Javascript, CSS, Python (Celery, Flask)**

DAIR Lab, Peking University

December 2025 – Oct 2026

Research Assistant

- Developed a self-evolving hyperparameter tuning framework, using a mixture of **Bayesian Optimization** and **open-source LLMs**
- Developed novel methods such as using a **Gaussian Process (GP)** to find similar tasks and **Functional Analysis of Variance (fanova)** to determine if the tuning job done on the task can be used.
- Designed a multi stage post training stack using **SFT, DPO** and **REINFORCE** algorithms, fine-tuning Qwen-3.5-9b, beating SOTA Bayesian Optimization algorithms and SOTA LLMs by **27%**.
- Extending transferability to **peptide and query optimization** by encoding combinatorial search space with a VAE into a continuous latent space.

Singapore Land Authority

March 2025 – June 2025

Data Engineer Intern

- Reverse-engineered legacy **geospatial logic** by deciphering and documenting complex “black-box” legacy algorithms for national land statistics to ensure data lineage and transparency.
- Architected **automated data pipelines** using custom ArcPy scripts to automate the ingestion and preprocessing of diverse geospatial datasets, achieving 99.8% parity with official land area metrics while maximizing output precision.
- Optimized computational efficiency by diagnosing and rectifying algorithmic inefficiencies in legacy ArcGIS workflows, implementing batch processing to stabilize national-scale vector data production.

PROJECTS

GEM-MM

- Built an **on-policy preference-alignment** method for vision-language models that scores **chain-of-thought** with a fork-versus-final entropy reward and updates the full VLM with **group-normalized policy gradients** (SEGA)
- Showed under a small budget of 3,000 MM-RLHF prompt pairs, standard multimodal DPO-style, image-conditional, and **self-improvement pair methods saturate**.
- On Qwen3-VL-4B, GEM-MM outperformed SOTA alignment methods under a small training budget (AMBER Hallucination, HallusionBench)

SKILLS

Technical

PyTorch, Scikit-Learn, Auto-Sklearn, Transformers,
ArcPy, Pandas, SQL

Languages

English (Native), Mandarin (Professional)